Why does sASM work — and only when the subdomain solve is inexact?

An interpretation through overlap over-counting and information propagation

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Minimal 1D / 2D / 3D Laplace experiments (all numerics in PETSc): BASIC vs. sASM additive Schwarz, with *exact* (Cholesky) and *inexact* (ICC(0)) subdomain solves.

BASIC = unscaled additive Schwarz; sASM = $\mathbf{D}^{-1/2}$-scaled (partition of unity). Full case list overleaf.

Case specifications

every experiment in this deck; all 2D/3D are constant-coefficient Laplace, BASIC/sASM additive Schwarz

Fig	Dim	Grid	Subdomains	O	Subdomain solver	Measures	Tol / k
1	1D	256 DOF	8	2, 4	exact Cholesky	one PC apply	—
2/10	1D	256 / 384	8 / 12	2/6/12	exact Cholesky	stationary front speed	—
3	2D	64^2	4×4	2	exact & ICC(0)	CG iterations	10^{-8}
4	2D	64^2	4×4	2	ICC(0)	residual field	k=5/15/30/60
5	2D	128^2	8×8	1/2/4	ICC(0)	residual field, iters	k=40
6	3D	n_x=48	4 (MPI ranks)	0/1/2	exact & ICC(0)	CG iterations	$10^{-8}/10^{-6}$
7	1D,2D	256 / 64^2	8 / 4×4	2	ICC(0), nat/RCM/rnd	iters, η, κ	10^{-8}
8	2D	120^2	6×6	1..16	exact & ICC(0)	$\lambda_{\max}, \lambda_{\min}$ (KSPCG)	eig. est.
9a	2D	120^2	6×6	2	ICC(0)	λ, κ, iters (1 vs 2-lvl)	10^{-8}
9b	2D	n=$20P$	P^2, P=4..12	2	ICC(0)	weak-scaling iters	10^{-8}

BC: 1D = left-node Dirichlet source; 2D = Dirichlet; 3D Sys 3 = 1 face Dirichlet + 5 faces Neumann. 1D ICC(0) = exact Cholesky (tridiagonal ⇒ no fill). Bound $\kappa \le C_0^2\,\omega\,N_c$, $\hat{N} = \max_k m_k \le N_c$.

sASM beats BASIC only with an inexact solve: 67 vs 102 (2D, ICC)

CG iterations to $\|r\|/\|b\| < 10^{-8}$; overlap $O{=}2$; only the subdomain solver changes

	1D Laplace (256, 8 sub)		**2D** Laplace (64^2, $4{\times}4$)	
	exact	ICC(0)	exact	ICC(0)
BASIC	15	15	23	**102**
sASM	25	25	27	**67**

- **sASM beats BASIC only in one cell: 2D with an inexact solve** (67 vs 102). Elsewhere sASM $\approx$ or slightly worse.
- 1D "ICC" $=$ exact *in natural ordering* (tridiagonal $\Rightarrow$ no fill), so 1D shows no anomaly *by default*. (Reorder the unknowns and 1D becomes inexact too — Fig. 7.)
- **Question:** what makes the *inexact* solve bad, and why does sASM fix it precisely there?

First, what is an "exact / inexact subdomain solve"?

$A_i = R_i A R_i^\top$ — **the global Laplace restricted to subdomain i (with overlap)**

- A_i implicitly imposes **homogeneous Dirichlet** on the *artificial cut* between subdomains (cross-interface coupling dropped), keeping the physical BC on faces touching the real boundary.
- **Exact solve** A_i^{-1} (Cholesky) $=$ *exactly* solving that local Dirichlet–Laplace subproblem. The inexactness factor $\omega = 1$.
- **Inexact solve** ICC(0) $=$ exact solve of a *perturbed* operator $\tilde{A}_i = A_i - E_i$, where E_i is the fill ICC(0) *drops*. This E_i **is** the source of $\omega > 1$, and its error sits in the overlap/seam.

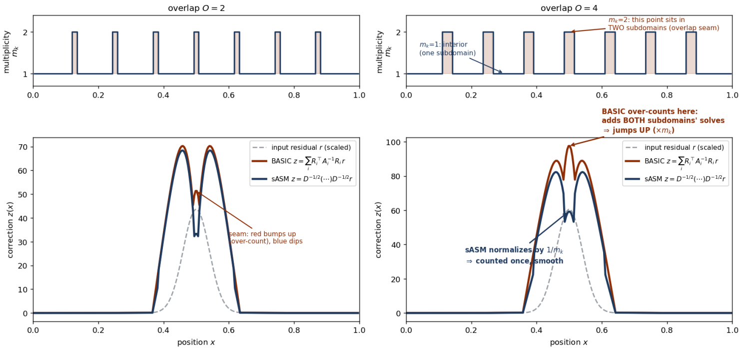
So "inexact" is not mysterious

ICC(0) keeps only A_i's nonzero pattern and throws away all fill-in. Whether that loses anything depends entirely on the **ordering** of the unknowns — demonstrated later by reordering (Fig. 7). The rigorous measure is $\omega = \lambda_{\max}(M_i^{-1} A_i)$, the local condition factor.

Mechanism: BASIC over-counts the overlap (the factor $\hat{N}$)

1D Laplace, 256 DOF, 8 subdomains; one apply to a localized residual — *both* curves are exact Cholesky

Fig. 1 (annotated) — BOTH lines are EXACT (Cholesky) solves; red BASIC over-counts the overlap, blue sASM normalizes it



There is NO 'inexact' line in this figure: in 1D, ICC(0) = exact Cholesky (a tridiagonal has no fill), so exact and ICC coincide. The exact-vs-inexact (ICC) contrast first appears in 2D — see Fig. 3.

- Over-count $= \sum_i R_i^\top R_i = \mathbf{D} = \mathrm{diag}(m_k)$, $\hat{N} = \max_k m_k$. **Harmless with an exact solve** (red ≈ blue); bites only once the solve is *inexact*. In 1D, ICC(0)=exact ⇒ no separate inexact curve here (that is 2D, Fig. 3).

Q1: inexactness bites because BASIC *multiplies* it by the over-count

the abstract additive-Schwarz bound: the two factors multiply in $\lambda_{\max}$

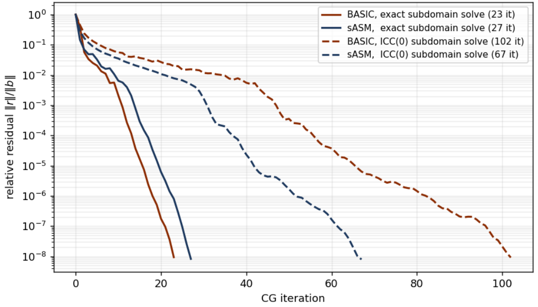
$$\kappa(\mathbf{M}^{-1}\mathbf{A}) \;\le\; C_0^2 \cdot \underbrace{\omega}_{\substack{\text{(A) inexactness}\\ \text{exact}=1,\ \text{ICC}>1}} \cdot \underbrace{N_{\mathrm{c}}}_{\substack{\text{(B) over-count}\\ \hat{N}=\max_k m_k \le N_{\mathrm{c}}}}$$

- Over-count factor: $\mathbf{D} = \mathrm{diag}(m_k)$ measures the **max multiplicity** $\hat{N}$; rigorously $\hat{N} \le N_{\mathrm{c}}$ (Gander et al. 2015, Rem. 2.8). It is what the scaling removes.
- **Exact** ($\omega{=}1$): corrections are exact; over-counting only lifts $\lambda_{\max} \le \hat{N}$ — bounded, CG tolerates it. *No anomaly* (23 vs 27).
- **ICC** ($\omega{>}1$): each correction carries an *error*; BASIC **adds** m_k **of them** in the overlap $\Rightarrow$ $\lambda_{\max} \approx \omega\,\hat{N}$, inexactness *times* over-count.
- Wider overlap: $\omega\uparrow$ *and* $\hat{N}\uparrow \Rightarrow \kappa$ *rises* with overlap (the anomaly); the amplified error **piles up at the seams**.

Q1 evidence: the over-counted ICC error stalls CG at the seams

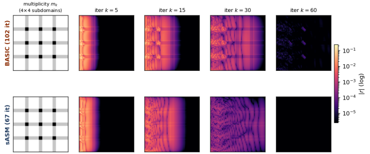
2D Laplace, 64^2, 4×4 subdomains, overlap $O{=}2$

Fig. 3 — 2D Laplace: sASM beats BASIC ONLY with an inexact solve
(exact: 23 vs 27; ICC(0): 67 vs 102) — over-counting bites only when coupled with inexac



Residual history (CG). Exact: BASIC≈sASM (both fast). ICC: BASIC **102** blows up, sASM **67**.

Fig. 4 — ICC(0) subdomain solve: residual |r| at MATCHED CG iterations. BASIC's residual lingers at the overlap seams; sASM clears it ⇒ 67 vs 102 iters



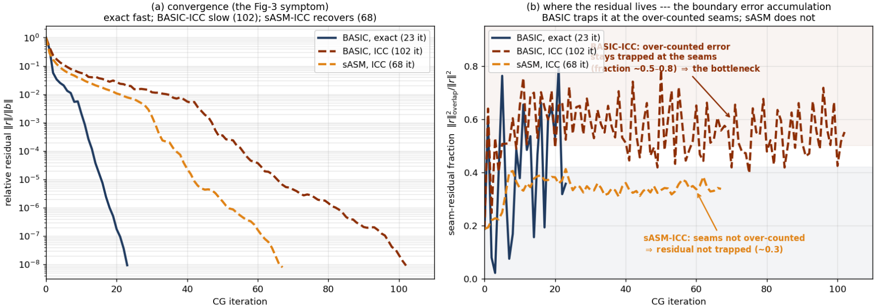
ICC residual field at matched CG iters $k{=}5/15/30/60$: at $k{=}60$ BASIC is still bright, the seams lit by the over-counted ICC error; sASM nearly dark.

"Inexact is bad" = the local error is **counted m_k times in the overlap** (A×B), accumulating into seam residual that costs iterations — and worsens with overlap.

Q1 evidence, localized: WHERE the inexact error accumulates

2D Laplace 64^2, 4×4, $O{=}2$; seam-residual fraction $= \|r\|^2_{\text{overlap}}/\|r\|^2$ per CG iteration

ı. 13 — Exact vs inexact at the seams: Fig 3 shows the convergence gap (a); the seam-residual fraction (b) shows WHERE the inexact error accumulat

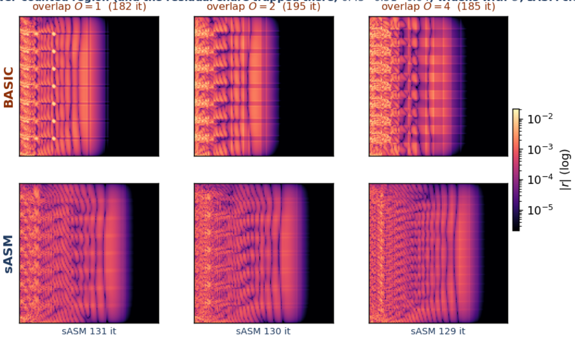


- **Fig. 3 (= panel a) shows only the global convergence gap** (exact 23, BASIC-ICC 102, sASM-ICC 68) — the *symptom*, not where it comes from.
- **Panel (b) localizes it:** BASIC traps the residual at the over-counted seams (fraction ∼0.5–0.8); with ICC that trapped error is the bottleneck. sASM does not over-count ⇒ seam fraction stays ∼0.3 ⇒ clears faster. *This* is the boundary error accumulation.

Q1, deeper: the seam error concentrates as overlap grows

2D Laplace, 128^2, 8×8 subdomains, ICC(0); residual $|r|$ at fixed iteration $k{=}40$, overlap $O{=}1, 2, 4$

Fig. 5 — Overlap sweep (128^2, 8×8, ICC(0), residual at fixed $k = 40$): BASIC's residual sits in the over-counted seams; the over-counted region (and the residual share trapped there, 0.45→0.51→0.64) widens with O; sASM clears it at every O

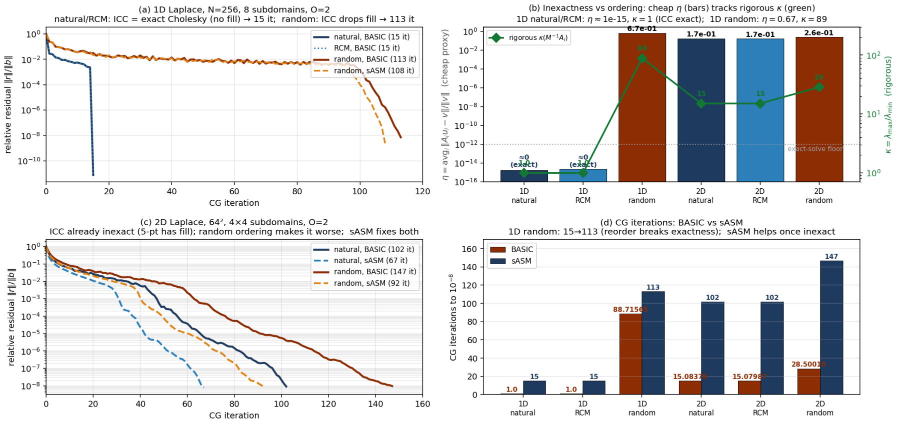


- The over-counted seam region *widens* with overlap, and the residual share trapped there grows $0.45 \to 0.51 \to 0.64$.
- sASM is better at *every* overlap (iters BASIC 182/195/185 vs sASM 131/130/129).
- Honest note: in 2D the iteration count vs O is non-monotone (ICC only mildly inexact in 2D). The clean monotone "overlap↑⇒iter↑" is a 3D effect → shown later (Fig. 6).

What “inexact” really is: reordering turns the ICC inexact

1D (256, 8 sub) and 2D (64^2, 4×4), $O{=}2$, ICC(0); only the ICC *ordering* changes (natural / RCM / random)

Fig. 7 — Reordering the subdomain unknowns turns the ICC factorization inexact (Q1: 1D natural ICC = exact; random ICC ≠ exact)



Only band-widening reorderings break exactness; sASM repairs them

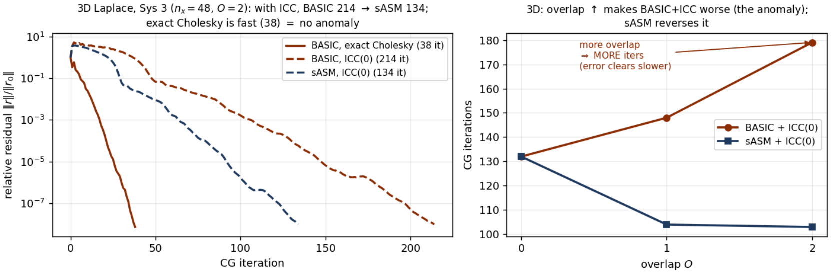
η (cheap ℓ_2 residual flag) and the rigorous $\kappa(M_i^{-1}A_i) = \lambda_{\max}/\lambda_{\min}$ track together; recall $\omega = \lambda_{\max}(M_i^{-1}A_i)$

- **(b) the headline.** 1D *natural & RCM*: $\eta \approx 10^{-15}$, $\kappa = 1$ — ICC *is* exact Cholesky (tridiagonal has *no fill*). 1D *random*: $\eta = 0.67$, $\kappa = 89$ — the permutation creates fill ICC(0) throws away, so ICC is now inexact. (η is only a flag; κ, the green line, is the rigorous condition number $= \omega$.)
- **(a)/(d) 1D.** natural/RCM: 15; random blows up to 113 — yet **sASM barely helps** (113→108), because 1D's over-count is only $\hat{N}{=}2$: A ($\kappa{=}89$) is large but B ($\hat{N}$) is small, so there is little to remove. (RCM stays banded ⇒ exact: not every permutation breaks it.)
- **(c)/(d) 2D.** The 5-point ICC already has fill ($\kappa{=}15$); random makes it worse ($\kappa{=}28$): $102 \to 147$ it. Here $\hat{N}{=}4$, so **sASM repairs both** ($102 \to 67$, $147 \to 92$). *The sASM win needs both A and B large* — 1D random (A only) vs 2D/3D.

On the real 3D solve (Sys 3): more overlap $\Rightarrow$ more BASIC+ICC iters; sASM reverses it

3D Laplace, 1 face Dirichlet + 5 faces Neumann, $n_x\!=\!48$, 4 MPI ranks — the actual torso-type solve

Fig. 6 — The same effect on the real 3D Sys 3 (mixed Dirichlet/Neumann Laplace)



- **Left (O=2, rtol 10^{-8}):** ICC — BASIC 214 slow, sASM 134 recovers; exact Cholesky 38 (no anomaly, the reference).
- **Right (overlap O=0/1/2, rtol 10^{-6}):** the clean monotone anomaly — BASIC+ICC $132 \to 148 \to 179$; sASM $132 \to 104 \to 103$ reverses it.

Q2: the $\mathrm{D}^{-1/2}$ partition of unity removes $\hat{N}$, so $\lambda_{\max} \le \omega$

$\mathbf{M}_{\mathrm{sASM}}^{-1} = \mathbf{D}^{-1/2}\big(\sum_i R_i^\top A_i^{-1} R_i\big)\mathbf{D}^{-1/2}$, partition of unity $\sum_i d_i^2 = 1$

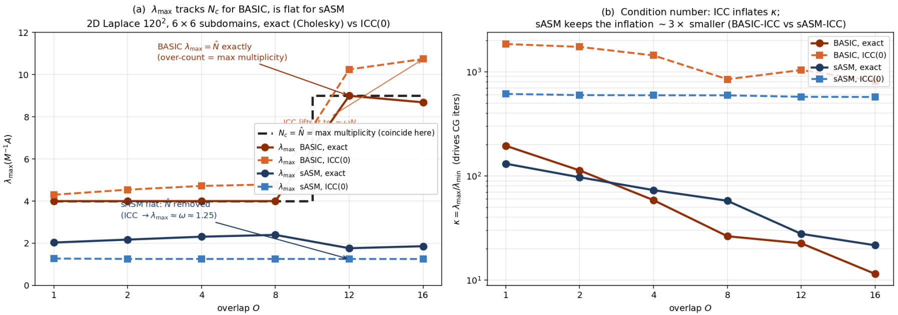
$$\text{BASIC: } \lambda_{\max} \le \omega\,\hat{N} \qquad \xrightarrow{\mathbf{D}^{-1/2}(\cdot)\mathbf{D}^{-1/2}} \qquad \text{sASM: } \lambda_{\max} \le \omega$$

- The $\mathbf{D}^{-1/2}$ sandwich restores a partition of unity ($\sum_i$ scaled $R_i^\top R_i = I$), so each DOF is counted **once**: the multiplicity over-count $\hat{N}$ is **removed** (Fig. 1 blue flat).
- **This breaks the multiplication.** ω is no longer multiplied by $\hat{N}$: the ICC error in the overlap is counted once, *not accumulated* $\Rightarrow$ no seam pile-up; CG recovers **102** $\to$ **67**.
- **sASM does not fix the inexactness** (ω still > 1) — it stops the over-count from *amplifying* it. So $\kappa \lesssim C_0^2\,\omega$, and overlap helps again. Measured next.

Measured directly: $\lambda_{\max}$(BASIC,exact) $= \hat{N}$, and sASM is flat

2D Laplace 120^2, $6{\times}6$ subdomains, overlap $O{=}1..16$; $\lambda_{\max}/\lambda_{\min}$ from KSPCG (PCSHELL wrapping BASIC/sASM)

Fig. 8 — Measuring the over-count: $\lambda_{\max}(\mathrm{BASIC,exact}) = \hat{N}$ (max multiplicity $\leq N_c$); sASM's $D^{-1/2}$ removes it ($\kappa \leq C_0^2\,\omega\,N_c$)

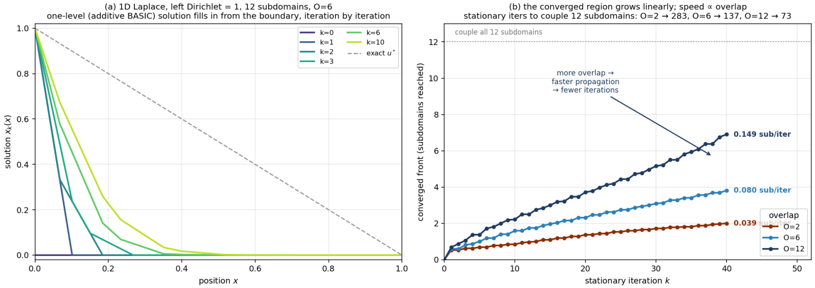


- **(a)** $\lambda_{\max}$(BASIC,exact) $= \hat{N}$ *exactly* (4, then 9 when overlap reaches the 2nd neighbour). ICC lifts it to $\approx \omega\hat{N}$. **sASM is flat** ($\lambda_{\max} \approx \omega = 1.25$ for ICC): $\hat{N}$ removed, the $\omega \times \hat{N}$ product broken.
- **(b)** ICC inflates κ $\sim$5–10$\times$; sASM-ICC keeps it $\sim$3$\times$ below BASIC-ICC $\Rightarrow\sim\sqrt{3}$ fewer CG iters. ($\hat{N} \leq N_c$; they coincide in this regular layout.)

Second thread: why one-level is slow at all — information propagation

1D Laplace, left-node Dirichlet, 12 subdomains (stride 32), *exact stationary* BASIC iteration — A sparse but A^{-1} dense

0 — Information propagation, quantified: one-level Schwarz couples the domain at finite speed (∝ overlap); that latency = the small $\lambda_{\min}$ a coarse space ren



- Each Schwarz step solves only *local* problems, so the converged front advances at **finite speed** (0.039/0.080/0.149 sub/iter for $O{=}2/6/12$) — additive, slower than the "1 subdomain/iter" multiplicative slogan.
- Cost to couple the domain **grows with #subdomains**, shrinks with overlap.

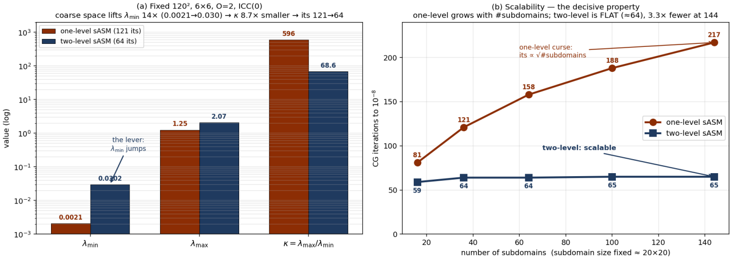
One thing, three views:
front speed (physical) $=$ small $\lambda_{\min}$ (spectral) $=$ no coarse level (algorithmic).
The next slide fixes all three.

Improving sASM: add a coarse space to lift $\lambda_{\min}$ (scalable)

sASM fixed $\lambda_{\max} \approx \omega$; the bottleneck is now tiny $\lambda_{\min} \geq C_0^{-2}$ (the propagation latency) $\Rightarrow$ two-level sASM

$$\mathbf{M}_2^{-1} = \underbrace{R_0^\top A_0^{-1} R_0}_{\text{coarse: lifts } \lambda_{\min}} + \underbrace{\mathbf{D}^{-1/2}\Big(\sum_i R_i^\top A_i^{-1} R_i\Big)\mathbf{D}^{-1/2}}_{\text{sASM fine: keeps } \lambda_{\max} \approx \omega} \quad \text{(Nicolaides PoU, symmetric, CG-OK)}$$

Fig. 9 — Improving sASM: add a coarse space. The bottleneck is $\lambda_{\min}$ (missing global coupling), not $\lambda_{\max}$; a two-level sASM fixes it and is scalable.

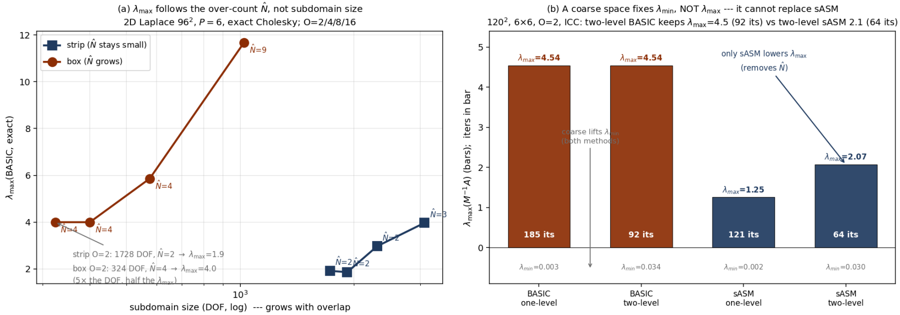


(a) fixed $120^2/6{\times}6/O{=}2$: $\lambda_{\min}$ 14× up, κ 8.7× down, its 121→64. (b) weak scaling: one-level →217; **two-level flat (≈64) = scalable**.

Controls: it is the over-count (not size), and a coarse space can't replace sASM

M^{-1} **verified fixed SPD (symmetry** 2.9×10^{-16}**,** $\lambda_{\min}>0$**)** $\Rightarrow$ **CG valid; outer stop** $\|r\|/\|b\|$ **uniform**

Fig. 11 — Controls: (a) the over-count $\hat{N}$ (not size) sets $\lambda_{\max}$; (b) a coarse space and sASM fix opposite ends of the spectrum --- you need both

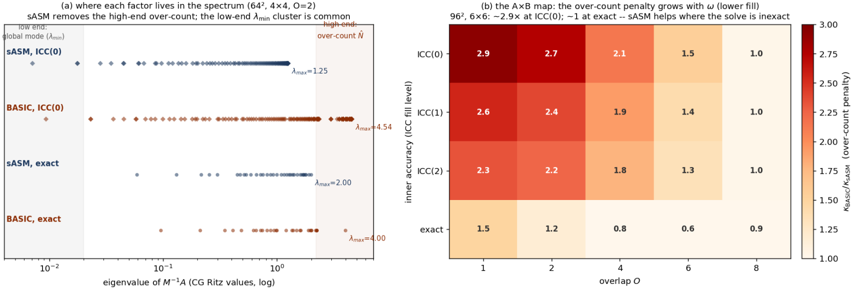


- **(a)** strip ($\hat{N}=2$, 1728 DOF) has *half* the $\lambda_{\max}$ of box ($\hat{N}=4$, 324 DOF) despite 5× the size: $\lambda_{\max}$ tracks $\hat{N}$, not subdomain size.
- **(b)** two-level BASIC keeps $\lambda_{\max}=4.5$ (92 its); only sASM lowers it to 2.1 (64 its). Coarse fixes $\lambda_{\min}$, sASM fixes $\lambda_{\max}$ — **orthogonal, both needed** (cost agrees: sASM-two 40 ms fastest).

Locating the mechanism: over-count at the high end; penalty $\propto$ inexactness

(a) full spectrum via CG Ritz values; (b) over-count penalty $\kappa_{\mathrm{BASIC}}/\kappa_{\mathrm{sASM}}$ over (overlap $\times$ ICC fill level)

Fig. 12 — Locating the mechanism: (a) over-count sits at the high end of the spectrum (sASM removes it); (b) its penalty scales with inner inexactness ω



	1	2	4	6	8
ICC(0)	2.9	2.7	2.1	1.5	1.0
ICC(1)	2.6	2.4	1.9	1.4	1.0
ICC(2)	2.3	2.2	1.8	1.3	1.0
exact	1.5	1.2	0.8	0.6	0.9

- **(a)** BASIC-ICC reaches $\lambda_{\max}=4.5$ (high-end outliers = over-count); sASM-ICC capped at 1.25. The low-end $\lambda_{\min}$ cluster is common — sASM acts at the *high* end, the coarse space at the *low* end.
- **(b)** the over-count penalty is $\sim$2.9$\times$ at ICC(0), fading to $\sim$1 at exact: it scales with ω — sASM pays off exactly where the solve is inexact.

Conclusion: over-count multiplies inexactness; the partition of unity breaks it

The one-sentence picture

BASIC counts the overlap m_k times (Fig. 1). With *exact* solves that is harmless — it only lifts $\lambda_{\max} \le \hat{N}$, which CG tolerates (Fig. 3, solid). With an *inexact* ICC solve the local error is counted m_k times, so "inexactness $\times$ over-count" ($\omega\hat{N}$) accumulates at the seams (Fig. 4) and convergence degrades (Fig. 3, dashed, 102). sASM's $\mathbf{D}^{-1/2}$ partition of unity removes the multiplicity over-count, breaking the product, so the inexact corrections are no longer amplified and CG recovers (102 $\to$ 67).

- **Q1:** inexactness is bad because BASIC *multiplies* it by $\hat{N} = \max_k m_k$; both grow with overlap (the anomaly). Measured: $\lambda_{\max}$(BASIC,exact) $= \hat{N}$, lifted to $\omega\hat{N}$ by ICC (Fig. 8).
- **Q2:** sASM removes $\hat{N}$ (partition of unity, Fig. 8 flat), so it helps *exactly* when the solve is inexact, and is neutral when exact.
- **Second thread:** information propagates at finite, overlap-set speed (Fig. 2/10) $=$ the small $\lambda_{\min}$; a symmetric **two-level sASM** (coarse space) lifts it, halving iterations and making them *scalable* (Fig. 9).